

OPERATIONS RESEARCH AND SYSTEMS ANALYSIS

Tutorial: Stochastic processes

Binomial distribution

- Discrete probability distribution
 - ▣ n independent yes/no experiments
 - ▣ k is number of successes
 - ▣ p is success probability
 - ▣ Then the probability of exactly k successes is as follows:

$$P(K = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

The number of ways
of picking k
unordered outcomes
from n possibilities.

$$\binom{n}{k} = \frac{n!}{k! (n - k)!}$$

Task 1

- 2nd credit test
- 10 multiple choice questions
 - ▣ 5 answers, exactly one is correct
- To obtain the credit, the student must mark at least 6 correct answers
- What is the probability that the student without any knowledge pass successfully the test only on the basis of random selection of answers?

Queuing theory – M/M/1

Average number of customers arriving in one unit of time	λ
The average number of customers who can be served per time period	μ
Server utilisation	$\rho = \lambda / \mu$
Average number of customers in the system.	$L = \rho / (1 - \rho)$
Average number of customers in the queue.	$L_Q = \rho^2 / (1 - \rho)$
Average time a customer spends in the system.	$W = 1 / (\mu - \lambda)$
Average time a customer spends in the queue.	$W_Q = \lambda / (\mu(\mu - \lambda))$
Probability that there are no customers in the system	$p_0 = 1 - \rho$

Task 2

- Elite EU Officer can sign 40 very important resolutions per hour in average.
 - In average, request for signing arrives every 2 minutes.
-
1. What is the average length of queue?
 2. What is the probability that the request will not wait?
 3. Interpret the value of server utilisation (too much, too low etc.)

Task 3

- There is one shop assistant in the flower shop. The service of one customer takes 5 minutes in average.
 - In average, 9 customers enter the shop per hour.
-
1. The goal is to reach the probability that a customer will not have to wait equal to 0.5. How long should be the average time to serve a customer?
 2. How many customers per hour should the assistant serve in average if the desired average time spent in the system is 10 minutes.

Markov chains

- Markov property:

- The probability of moving from state i at stage n to state j at stage $n + 1$ depend only on state at stage n and transition probabilities.

$$P\{X_{n+1}=i_{n+1} \mid X_0=i_0, \dots, X_n=i_n\}=P\{X_{n+1}=i_{n+1} \mid X_n=i_n\}$$

- Transition probability p_{ij}

- the probability that the process moves from state i at one stage to state j at the next stage.

Transition Matrix $\mathbf{P}$

- Always square matrix
- Sum in the row is always 1

$$\mathbf{P} = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1n} \\ p_{21} & p_{22} & \cdots & p_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ p_{n1} & p_{n2} & \cdots & p_{nn} \end{bmatrix}$$

- We will work just with time-homogenous markov chains

State vector

- $p(0)$ is initial state distribution

$$p^T(1) = p^T(0)\mathbf{P}$$

$$p^T(2) = p^T(1)\mathbf{P}$$

$$\vdots$$

$$p^T(n) = p^T(n-1)\mathbf{P}$$

Task 4

- The weather forecast can be classified as Sunny (S), Cloudy (C) or Rainy (R).
- If it is sunny today, the probability of sunny day tomorrow is 0.7 and cloudy day 0.1. If it is cloudy then it will be sunny with the probability of 0.5 and cloudy with 0.25.
- If it rains today it will be sunny tomorrow (0.4) and rainy again with (0.3).
- Define a transition matrix.
- Calculate the probability of a rainy day for the day after tomorrow if it is rainy today.

Task 5

- Three water lilies on pool: **A, B, C**.
 - A frog sits on the water lili **A**.
 - A hungry but lazy stork waits by the water lili **C**. (Stork eats frogs)
 - The bell in near church rings each hour..
 - The bell scares the frog. If the bell rings the frog randomly jumps to another lili (50:50)
-
1. Define a transition matrix.
 2. Calculate the probability that the stork is still hungry at 12:01 subject to the fact that the frog sits on the water lili **A** at 7:59.

Thank you for your attention

